



CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

Welcome back to the CodeChef Puzzle Challenge sponsored by *LINUX for You* and Directi. It's time to tease your brain with another interesting problem, and while you're at it, you may just win some cash as well!

Last month's puzzle

A proper divisor of a number is a divisor that is less than the number itself. Find the sum of all proper divisors for the number 1,23,748. For example, the proper divisors for the number 10 are 1, 2 and 5. So the answer would be $1+2+5 = 8$.

The solution

The sum of all the proper divisors for the number 1,23,748 is 92,818.

How did we do it?

A simple solution would be to run a for loop from 1 to 1,23,747 and if the

current number under consideration divides 1,23,748, then add it to the sum. If we have to find the divisor-sum for a lot of numbers, a more advanced technique like sieving could be used to pre-compute the answer.

And the winners for January are...

- Neeraj Kumar
- Debanjan Chanda
- Osric Fernandes

Puzzle for February

The Fibonacci number series is a famous one where $F[n] = F[n-1] + F[n-2]$. The first few numbers of this series are 0, 1, 1, 2, 3, 5, 8, and so on. Considering 1-based indices, what is the 999th Fibonacci number? (For example, The 7th Fibonacci number is 8).

Looking for a little extra cash?

Then solve this month's CodeChef Challenge puzzle and e-mail your answer to codechef@efyindia.com by February 18, 2010. Three lucky winners will win cash of Rs 1,000 each!

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. CodeChef has been created by Directi, as a way to continuously challenge and engage the developer community. The goal is to provide a platform for practice, competition and improvement as well as enable developers to benchmark their skills against their peers. Log on to www.codechef.com for the February Challenge that takes place from February 1 to 11, 2010, to win cash prizes up to Rs 55,000.